

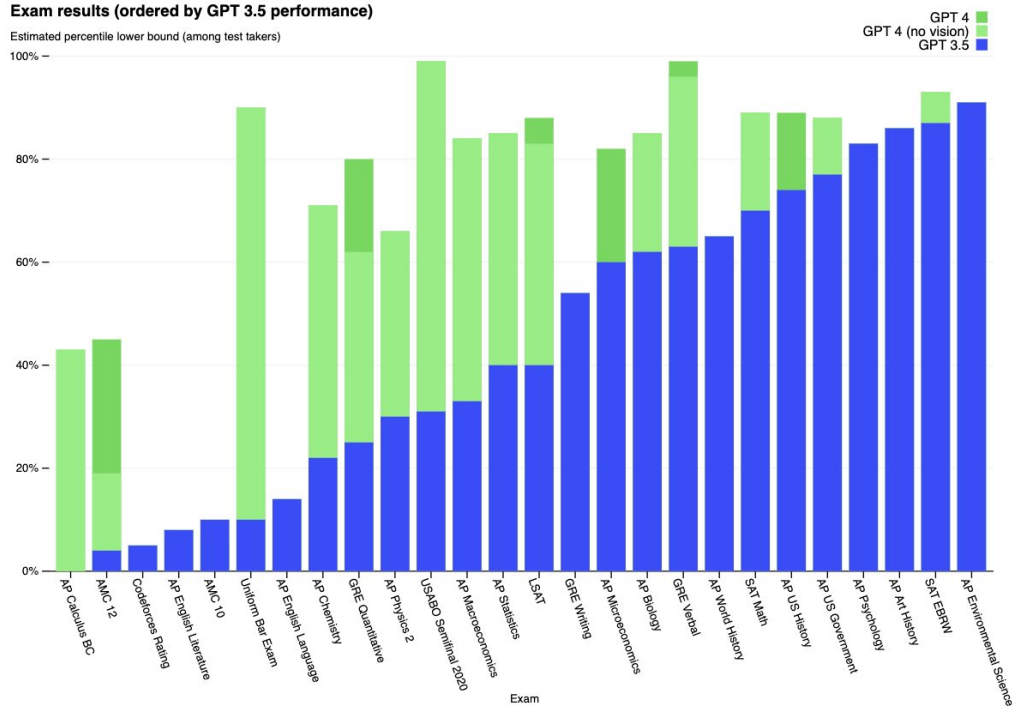


Figure 1: Taken directly from GPT-4 Technical Report (OpenAI, 2023b). To get a sense of how quickly model capabilities are progressing – consider the jump in exam performance between GPT-3.5 and GPT-4 (OpenAI, 2023b).

Our study is motivated less by the progress of these models alone though, and more by the breadth, scale, and capabilities we’ve seen in the complementary technologies developed around them. The role of complementary technologies remains to be seen, but maximizing the impact of LLMs appears contingent on integrating them with larger systems (Bresnahan, 2019; Agrawal et al., 2021). While the focus of our discussion is primarily on the generative capabilities of LLMs, it is important to note that these models can also be utilized for various tasks beyond text generation. For example, embeddings from LLMs can be used for custom search applications, and LLMs can perform tasks such as summarization and classification where the context may be largely contained in the prompt.

To complement predictions of technology’s impacts on work and provide a framework for understanding the evolving landscape of language models and their associated technologies, we propose a new rubric for assessing LLM capabilities and their potential effects on jobs. This rubric (A.1) measures the overall exposure of tasks to LLMs, following the spirit of prior work on quantifying exposure to machine learning (Brynjolfsson et al., 2018; Felten et al., 2018; Webb, 2020). We define exposure as a proxy for potential economic impact without distinguishing between labor-augmenting or labor-displacing effects. We employ human annotators and GPT-4 itself as a classifier to apply this rubric to occupational data in the U.S. economy, primarily sourced from the O\*NET database.<sup>12</sup>

To construct our primary exposure dataset, we collected both human annotations and GPT-4 classifications, using a prompt tuned for agreement with a sample of labels from the authors. We observe similar agreement

<sup>1</sup>This is distinct from recent social science research that makes use of LLMs to simulate human behavior (Horton, 2023; Sorensen et al., 2022)

<sup>2</sup>While our exposure rubric does not necessarily tie the concept of language models to any particular model, we were strongly motivated by our observed capabilities of GPT-4 and the suite of capabilities we saw in development with OpenAI’s launch partners (OpenAI, 2023b).

levels in GPT-4 responses and between human and machine evaluations, when aggregated to the task level. This exposure measure reflects an estimate of the technical capacity to make human labor more efficient; however, social, economic, regulatory, and other determinants imply that technical feasibility does not guarantee labor productivity or automation outcomes. Our analysis indicates that approximately 19% of jobs have at least 50% of their tasks exposed when considering both current model capabilities and anticipated tools built upon them. Human assessments suggest that only 3% of U.S. workers have over half of their tasks exposed to LLMs when considering existing language and code capabilities without additional software or modalities. Accounting for other generative models and complementary technologies, our human estimates indicate that up to 49% of workers could have half or more of their tasks exposed to LLMs.

Our findings consistently show across both human and GPT-4 annotations that most occupations exhibit some degree of exposure to LLMs, with varying exposure levels across different types of work. Occupations with higher wages generally present with higher exposure, a result contrary to similar evaluations of overall exposure to machine learning (Brynjolfsson et al., 2023). When regressing exposure measures on skillsets using O\*NET's skill rubric, we discover that roles heavily reliant on science and critical thinking skills show a negative correlation with exposure, while programming and writing skills are positively associated with LLM exposure. Following Autor et al. (2022a), we examine barriers to entry by "Job Zones" and find that occupational exposure to LLMs weakly increases with the difficulty of job preparation. In other words, workers facing higher (lower) barriers to entry in their jobs tend to experience more (less) exposure to LLMs.

We further compare our measurements to previous efforts documenting the distribution of automation exposure in the economy and find broadly consistent results. Most other technology exposure measures we examine are statistically significantly correlated with our preferred exposure measure, while measures of manual routineness and robotics exposure show negative correlations. The variance explained by these earlier efforts (Acemoglu and Autor, 2011a; Frey and Osborne, 2017; Brynjolfsson et al., 2018; Felten et al., 2018; Webb, 2020; Brynjolfsson et al., 2023), along with wage controls, ranges from 60 to 72%, indicating that 28 to 40% of the variation in our AI exposure measure remains unaccounted for by previous technology exposure measurements.

We analyze exposure by industry and discover that information processing industries (4-digit NAICS) exhibit high exposure, while manufacturing, agriculture, and mining demonstrate lower exposure. The connection between productivity growth in the past decade and overall LLM exposure appears weak, suggesting a potential optimistic case that future productivity gains from LLMs may not exacerbate possible cost disease effects (Baumol, 2012; Aghion et al., 2018).<sup>3</sup>

Our analysis indicates that the impacts of LLMs like GPT-4, are likely to be pervasive. While LLMs have consistently improved in capabilities over time, their growing economic effect is expected to persist and increase even if we halt the development of new capabilities today. We also find that the potential impact of LLMs expands significantly when we take into account the development of complementary technologies. Collectively, these characteristics imply that Generative Pre-trained Transformers (GPTs) are general-purpose technologies (GPTs).<sup>4</sup> (Bresnahan and Trajtenberg, 1995; Lipsey et al., 2005).

(Goldfarb et al., 2023) argue that machine learning as a broad category is likely a general-purpose technology. Our evidence supports a wider impact, as even subsets of machine learning software meet the criteria for general-purpose technology status independently. This paper's primary contributions are to provide a set of measurements of LLM impact potential and to demonstrate the use case of applying LLMs to develop such measurements efficiently and at scale. Additionally, we showcase the general-purpose potential of LLMs. If "GPTs are GPTs," the eventual trajectory of LLM development and application may be challenging for

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<sup>3</sup>Baumol's cost disease is a theory that explains why the cost of labor-intensive services, such as healthcare and education, increases over time. This happens because wages for skilled workers in other industries increase, but there is no corresponding increase in productivity or efficiency in these service industries. Therefore, the cost of labor in these industries becomes relatively more expensive compared to other goods and services in the economy.

<sup>4</sup>For the remainder of the paper we spell out general-purpose technologies when it is used outside of stating "GPTs are GPTs."